

Reference-based Dense Pose Estimation via Partial 3D Point Cloud Matching

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ABSTRACT

Interacting with real-world objects is one of the fundamental tasks in multimedia. Despite its importance, existing object pose estimation targets only rigid objects. This demonstration proposes a novel application for non-rigid object pose estimation. Inspired by human dense pose estimation, we represent a pose of a non-rigid object as an indexed point cloud, where each index corresponds to that in a template. The correspondence is identified by a machine-learning-based 3D point cloud matching. Finding correspondence to the template point cloud enables a dense pose estimation with no object-specific learning processes. In the demonstration, we visualize the correspondence of points in observed depth images and the template. We also provide a demonstration of template point cloud reconstruction. Through these systems, onsite visitors can test our system with objects brought by themselves and have an experience with a state-of-the-art 3D point cloud matching method as well as this novel task.

CCS CONCEPTS

• **Computing methodologies** → *Graphics processors; Matching; Shape inference; Tracking; Vision for robotics; Scene understanding.*

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1 INTRODUCTION

We demonstrate a novel task of *reference-based 3D dense pose estimation* that estimates the 3D dense pose of arbitrary non-rigid objects (see Fig. 1). When a system interacts with real-world objects through whatever modalities (e.g., physically or optically),

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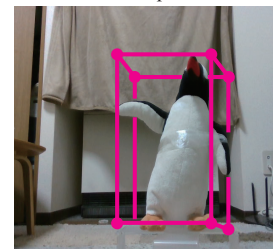
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Conventional 3D pose estimation



Ours

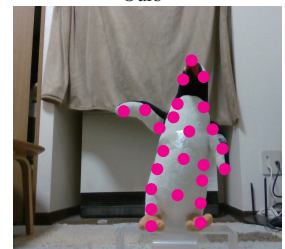


Figure 1: The concept of our system.

it is crucial to identify the location of surface points referring to its template [2, 10, 12]. Dense pose estimation for specific objects such as human body and human hand is known for that [1, 3, 5], but it requires model training for the target object of humans. In our concept, we achieve similar dense pose estimation but without learning for each target object.

To realize the concept, we propose a novel application of 3D point cloud matching, which is a problem to find correspondence between source and target point clouds [6–8]. In our application, the source is the template 3D point cloud and the target is a single-frame observation (e.g., depth image). When objects are rigid and their 3D shape is known, we can trivially convert a 6D pose into the dense pose [9, 10]. In contrast, the 6D pose is insufficient or undefinable for non-rigid objects due to deformation. Our application defines a 3D pose of a non-rigid object as an indexed point cloud. Here, each index is tied with that of the template point cloud.

The conventional 6D pose estimation is fundamental for multimedia systems that involve humans, robots, and projectors interacting with rigid objects. Our application extends that fundamental technique for non-rigid objects. Thus, it would similarly impact the multimedia community. In the demonstration, we visualize the correspondence of point indices between the reference and 2.5D depth images. Visitors can test the system with multiple types of non-rigid objects with different poses and camera angles. We will also provide the functionality to reconstruct a full 3D point cloud as the template. Then, onsite visitors can even test with objects brought by themselves.

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Figure 2: The hardware items of our demonstration.

2 HARDWARE SETTING

Figure 2 shows the hardware setting of our demo system. It is organized by two items: a depth camera (Intel Realsense Depth Camera D435) and a Laptop (ELSA VELUGA G4-NM 17A55M). The laptop PC has Intel Core i9-12900H as its CPU, NVIDIA RTX A5500 (Laptop) as its GPU, and 32GB×2 main memory.

The depth camera captures non-rigid objects as time series data. Although estimation does not run in real-time, dense pose estimation is applied online for captured frames independently.

3 IMPLEMENTATION

This section explains our partial 3D point cloud matching strategy, foreground object extraction with depth images, reference point cloud reconstruction from multiple depth images, and visualization.

3.1 Partial 3D point cloud matching

Recent progress in 3D point cloud matching is drastic. Some methods even achieved the challenging *partial 3D point cloud matching* with non-rigid objects [4, 11] in a certain accuracy. 4DMatch and 4DLoMatch are the benchmark task of that problem; 4DMatch is 3D point cloud matching with non-rigid deformation, and 4DLoMatch is a low overlap version of 4DMatch.

The concept of our task, reference-based dense pose estimation, has a difference from 4DMatch in its input. 4DMatch assumes that both source and target point clouds have missing parts (partial-partial matching). In contrast, our source point cloud, the template, is a reconstructed full 3D point cloud, and only the target has missing parts (full-partial matching). In this sense, our task is essentially more feasible than 4DMatch or 4DLoMatch benchmarks. Nonetheless, our task is the first practical application of non-rigid 3D point cloud matching.

In this demonstration, we use our previous work, edge-selective feature weaving module (ESFW) [11] combined with Leopard, a feature extractor proposed in [4]. Recent 3D point cloud matching methods consist of two modules: a feature extractor and a correspondence identifier. Leopard proposed a novel feature extractor, while ESFW is a novel correspondence identifier. Using ESFW instead of conventional correspondence identifiers, our implementation boosts the performance of Leopard and gives a state-of-the-art performance [11].

The model is trained on the 4DMatch dataset following [11] and [4]. Note that our system uses PointMLP [6] instead of KPFCN [7] as the feature extraction module for computational efficiency and matching performance.

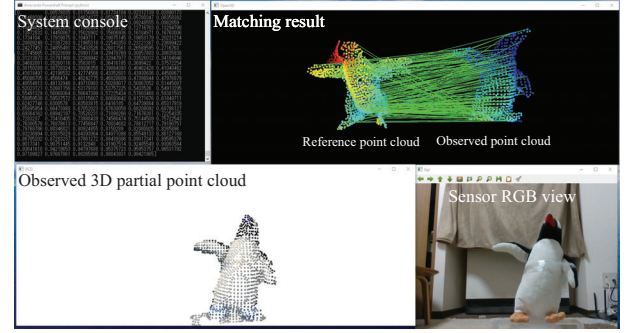


Figure 3: The visual of our demo system. The 2.5D point cloud is matched to pre-registered 3D point cloud even under the target object shape deformation.

3.2 Foreground object extraction in depth image

To extract the foreground point cloud, we assume that the background point cloud is far enough compared with the foregrounds and remove the point cloud that is farther than a depth threshold in the camera coordinate system. Here, point clouds with smaller depths from the threshold are extracted as the foreground. In our setting, only the points whose z-coordinate value is smaller than 1m are taken as the foreground.

3.3 Reference point cloud reconstruction from multiple depth images

We create a complete 3D reference point cloud by combining partial-view point clouds from several viewpoints. While making a reference, we assume the measuring object does not deform. We use the matching method described in Subsec. 3.1 for matching estimation and then extract the 3D rigid transform from the matching. After that, the combined point cloud was downsampled by voxel grid downsampling to obtain a spatially uniform reference point cloud. To provide better matching when creating the reference, the viewpoints are not positioned extremely far from each other, and the target object is kept not deforming.

3.4 Visualization

The constructed reference point cloud is then used for visualizing 3D dense pose estimation results. Similar to the reference construction, our system estimates matching between the reference and the test point cloud. Note that the test point cloud can be deformable. The matching results are visualized so that the corresponding points are the same color as in Fig. 3. Users can see the estimation results from various views by manipulating the camera using the mouse (Top right window in Fig. 3).

4 CONCLUSION

This paper proposed a novel task of reference-based dense pose estimation with a solution based on partial 3D point cloud matching. Before this study, human dense pose estimation is the only known strategy of the pose estimation task for non-rigid real 3D objects. In our implementation, Leopard [4] with ESWN [11] is adopted as the partial 3D point cloud matching solver, while we provided a system to reconstruct full 3D point cloud, which can be used as the template.

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